



King Abdulaziz University

Faculty of Computing & Information Technology Computer Science Department

Exam 1 (February 13th, 2019)
Course: CPCS-223 (Analysis & Design of Algorithms)
Term: Spring 2019 (Semester 2, 1439/1440)
Points: 60 Time: 80 minutes

SOLUTION

Student ID: _____

Student Name: _____

***Section: _____

CLO	SO	Question #
02	A	2
04	B	3
05	B	3, 5, 6
15	C	4

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MCQs

Question No. 1

05 points

Every Correct Answer will have 0.5 points

1. For array A, |A| is:

- a. the absolute value of the sum of A's elements;
- b. the absolute value of A;
- c. the smallest element of A;
- d. the number of elements in A;**
- e. None of these

Answer d.

2. How many nodes we may require to visit while searching of a linked list of n nodes?

- a. 1;
- b. 2;
- c. n;**
- d. n^2 ;
- e. none of these

Answer c.

3. Function “g” is a upper bound on function “f” iff for all x

- a. $g(x) \leq f(x)$;
- b. $g(x) \geq f(x)$;
- c. $g = O(f)$;
- d. $f = \Omega(g)$;
- e. none of these**

Answer e.

4. Best case for an algorithm:

- a. takes the same time for all data;
- b. assumes the data that the algorithm handles in the greatest time;
- c. assumes the data that the algorithm handles in the least time;**
- d. is the expected time considering all possible input data;
- e. none of these

Answer c.

5. Graphs are implemented as:

- a. trees;
- b. matrices or arrays of lists;**
- c. queues;
- d. stacks;
- e. none of these

Answer b.

6. The minimum number of edges in “a connected acyclic” graph on N vertices is

- a. **N-1**
- b. N
- c. N+1
- d. None of these

Answer a.

7. The number of key comparisons made “in the best case by insertion sort” for an array of size 10 is

- a. 90
- b. 45
- c. 50
- d. 30
- e. **None of the above**

Answer e.

8. What is the time efficiency of BFS on a graph of n vertices and e edges?

- a. **depends on graph implementation**
- b. $\Theta(n^2)$
- c. $O(n + e)$
- d. $\Theta(n)$
- e. $O(e)$

Answer a.

9. Matrix multiplication by brute force requires:

- a. $O(n)$ time;
- b. $O(1)$ time;
- c. $O(\log n)$ time;
- d. **nested loops**
- e. recursion

Answer d.

10. Linked lists are:

- a. static;
- b. branching;
- c. **linear;**
- d. stack allocated;
- e. none of these

Answer c.

CLO – 2
Question No. 2

SO – A
10 points

For each of the following pairs of functions, indicate whether the first function of each of the following pairs has a lower, same, or higher order of growth (to within a constant multiple) than the second function.

Every Correct Answer will have 1.5 points and 1 point extra if all correct

- | | |
|---|---|
| a. $n(n + 1)$ and $2000n^2$ a. same | b. $100n^2$ and $0.01n^3$ b. lower |
| c. $\log_2 n$ and $\ln n$ c.(higher) | d. $\log_2^2 n$ and $\log_2 n^2$ d.lower |
| e. 2^{n-1} and 2^n c. same | f. $(n - 1)!$ and $n!$ f. same |

OR

Prove that $a n^2 + b n + c = \Theta (n^2)$ **easy according to the limit of $a n^2 + b n + c$ when n goes to the infinity**

Solution:

$$a n^2 + b n + c = \Theta (n^2)$$

$$f(n) = a n^2 + b n + c$$

and

$$g(n) = \Theta (n^2)$$

we have to prove

$$c_2 g(n) \leq f(n) \leq c_1 g(n)$$

L. H. S (05 pts)

R. H. S (05 pts)

$$c_2 (n^2) \leq a n^2 + b n + c$$

$$a n^2 + b n + c \leq c_1 n^2$$

$$c_2 \leq a + b/n + c/n^2$$

$$a + b/n + c/n^2 \leq c_1$$

for $n = \infty$

for $n = 1$

this expression is true for all $n \geq 1$

this expression is true for all $n \geq 1$

and

and

$$c_2 \leq a$$

$$c_1 \geq a + b + c$$

*Question No. 3**10 points*

Consider the following recursive algorithm.

ALGORITHM *Riddle*($A[0..n - 1]$)
 //Input: An array $A[0..n - 1]$ of real numbers
if $n = 1$ **return** $A[0]$
 else $temp \leftarrow Riddle(A[0..n - 2])$
if $temp \leq A[n - 1]$ **return** $temp$
 else return $A[n - 1]$

- What does this algorithm compute? **It returns the min element of the input array (02 pts)**
- What is the basic operation of this algorithm? **The comparison (02 pts)**
- Set up a recurrence relation for the algorithm's basic operation count and solve it. **The recurrence relation is: $Comp(n) = Comp(n-1) + 1$ after that it should be solved**

Solution:

By backward substitution method

(06 pts)

$$\begin{aligned} Comp(n) &= Comp(n-1) + 1 \\ &= Comp(n-2) + 2 \\ &= Comp(n-3) + 3 \end{aligned}$$

⋮

$$Comp(n) = Comp(n-1) + 1 = Comp(n-2) + 2 = Comp(n-3) + 3 = \dots = Comp(n-i) + i$$

$$\begin{aligned} \text{For } n = i \\ &= n \end{aligned}$$

CLO – 15

SO – C

Question No. 4

15 points

Design a recursive algorithm for computing 2^n for any nonnegative integer n that is based on the formula $2^n = 2^{n-1} + 2^{n-1}$.

Then give the corresponding complexity

Note: the design can be provided as a pseudocode or as explicit and clear steps.

Solution:

Tow_Powern(n)

(05 pts)

```
{  
    If n=0 return (1)  
    Else return Tow_Powern(n-1) + Tow_Powern(n-1)  
}
```

The corresponding complexity is 2^n

Solution:

$$T(n) = T(n-1) + T(n-1) + 1$$

1

$$T(n) = 2 T(n-1) + 1$$

By backward substitution

$$n = n - 1 \quad T(n-1) = 2 T(n-2) + 1$$

$$2 \quad T(n) = 2 [2 T(n-2) + 1] + 1$$

$$T(n) = 4 T(n-2) + 3$$

$$n = n - 2 \quad T(n-2) = 2 T(n-3) + 1$$

$$3 \quad T(n) = 4 [2 T(n-3) + 1] + 3$$

$$T(n) = 8 T(n-3) + 7$$

$$n = n-3 \quad T(n-3) = 2 T(n-4) + 1$$

$$4 \quad T(n) = 8 [2 T(n-4) + 1] + 7$$

$$T(n) = 16 T(n-4) + 15$$

(05 pts)

So for “i th” call

$$T(n) = 2^i T(n - i) + 2^i - 1$$

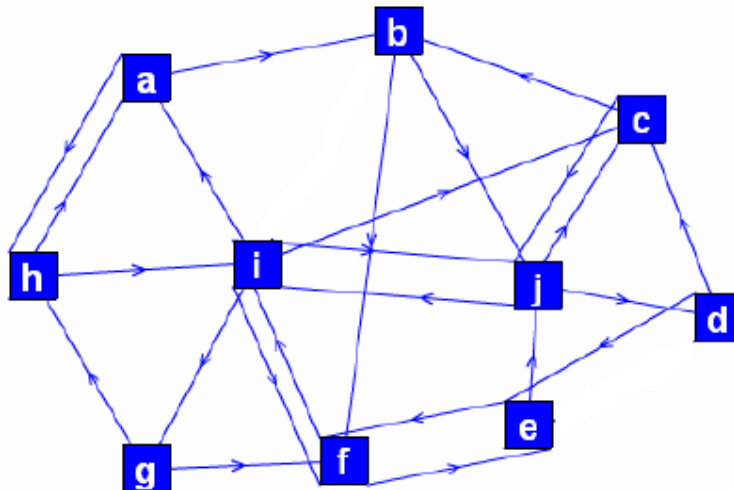
(03 pts)

This recursion will stop as $n - i = 0$, that implies $n = i$

$$\begin{aligned} T(n) &= 2^n T(n - n) + 2^n - 1 \\ &= 2^n T(0) + 2^n - 1 \\ &= 2^n \cdot 1 + 2^n - 1 \\ &= 2^n + 2^n - 1 \\ &= 2^{n+1} - 1 \end{aligned}$$

(02 pts)

- A. Give the adjacency matrix representation of the following graph
 B. Give the adjacency list representation of the following graph
 C. Apply DFS (Depth First Search) then BFS (Breadth First Search) on the following digraph starting from “a”.

**Solution:**

Adjacency Matrix it is a matrix of size 10X10 where 10 is the total number of vertices each cell includes 0 if there is no arrow from row vertex to column vertex and 1 otherwise.

Adjacency linked list: a is adjacent to b and h, b is adjacent to f and j and so forth as you know

DFS: a, b, f, e, j, c, d, i, g, h

(03 pts)

BFS: a, b, h, f, j, i, e, c, d, g

(03 pts)

Adjacency Matrix (02 pts)

	a	b	c	d	e	f	g	h	i	j
a	0	1	0	0	0	0	0	1	0	0
b	0	0	0	0	0	1	0	0	0	1
c	0	1	0	0	0	0	0	0	0	1
d	0	0	1	0	1	0	0	0	0	0
e	0	0	0	0	0	1	0	0	0	1
f	0	0	0	0	1	0	0	0	1	0
g	0	0	0	0	0	1	0	1	0	0
h	1	0	0	0	0	0	0	0	1	0
i	1	0	1	0	0	1	1	0	0	1
j	0	0	1	1	0	0	0	0	1	0

Adjacency list (02 pts)

a	→	b, h
b	→	f, j
c	→	b, j
d	→	c, e
e	→	f, j
f	→	e, i
g	→	f, h
h	→	a, i
i	→	a, c, f, g, j
j	→	c, d, i

CLO – 5

SO – B

Question No. 6

10 points

Determine the number of character comparisons made by the brute-force algorithm in searching for the pattern GANDHI in the following text:

THERE_IS_MORE_TO_LIFE_THAN_INCREASING_ITS_SPEED

Assume that the length of the text—it is 47 characters long—is known before the search starts.

Solution: 43

Text Length = $n = 47$

Pattern Length = $m = 6$

of Comparisons = $m - n + 1$

So

of Comparisons = $47 - 6 + 1 = 42$ in my opinion the correct answer is 42

```
ALGORITHM BruteForceStringMatch( $T[0..n - 1]$ ,  $P[0..m - 1]$ )
//Implements brute-force string matching
//Input: An array  $T[0..n - 1]$  of  $n$  characters representing a text and
//       an array  $P[0..m - 1]$  of  $m$  characters representing a pattern
//Output: The index of the first character in the text that starts a
//        matching substring or  $-1$  if the search is unsuccessful
for  $i \leftarrow 0$  to  $n - m$  do
     $j \leftarrow 0$ 
    while  $j < m$  and  $P[j] = T[i + j]$  do
         $j \leftarrow j + 1$ 
    if  $j = m$  return  $i$ 
return  $-1$ 
```

Extra Sheet for rough work